Astrophysics

Mark scheme and worked solutions for 25 questions: 8 multiple choice, 17 structured.

1 · Part 1

Paper 1 & 2 Units [1]

Answer: C

Worked solution

Mass (kg) is the amount of matter in an object. Weight (N) is the gravitational force on that mass. Gravitational field strength (N/kg) is the force on each kilogram, and weight = mass × gravitational field strength.

2 · Part 1

Paper 1 & 2 Motion in the universe [1]

Answer: A

Explanation

The universe is a large collection of billions of galaxies. B greatly underestimates the number of galaxies. C confuses our galaxy, the Milky Way, with the whole universe. D is wrong because the universe contains billions of galaxies, each made of billions of stars.

3 · Part 1

Paper 1 & 2 Motion in the universe [1]

Answer: B

Explanation

Gravitational force makes moons orbit planets, artificial satellites orbit the Earth and comets orbit the Sun. A and D wrongly have a moon orbiting the Sun directly. C and D wrongly have an artificial satellite orbiting the Moon and a comet orbiting a planet.

4 · Part 1

Paper 1 & 2 Stellar evolution [1]

Answer: D

Worked solution

In order of increasing surface temperature, star colours go red, orange, yellow, white, blue.

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5 · Part 1 Paper 2 only Stellar evolution [1]

Answer: B

Worked solution

The magnitude scale runs "backwards": a smaller number means a brighter star, and negative values are brighter still. Order of the stars from brightest to dimmest: Q (−1.5), S (+0.6), P (+4.8), R (+10.2).

6 · Part 1 Paper 2 only Cosmology [1]

Answer: A

Explanation

Light from more distant galaxies shows a greater red-shift. Galaxy J has the largest value of $\Delta\lambda / \lambda 0$ (0.031), so it is the furthest away. C (galaxy L) has the smallest red-shift, so it is the nearest. B and D have red-shifts smaller than J's, so they are closer than J.

7 · Part 1 Paper 2 only Cosmology [1]

Answer: D

Explanation

When a source moves towards an observer, the waves reaching the observer are squashed together. The observed wavelength is shorter and the observed frequency is higher (a blue-shift). A describes a source moving away (red-shift). B and C are impossible combinations, because for light a longer wavelength always goes with a lower frequency.

8 · Part 1 Paper 2 only Cosmology [1]

Answer: C

Worked solution

Rearrange the equation: velocity of galaxy = (change in wavelength / reference wavelength) × speed of light = $0.020 \times 3.0 \times 10^8$ m/s = 6.0×10^6 m/s. Check: $(6.0 \times 10^6) / (3.0 \times 10^8) = 0.020$.

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9 · Part 1

Paper 1 & 2 | Motion in the universe | [7]

Mark scheme

(a) **[3]**

- weight = mass × gravitational field strength OR W = m × g (1)
- W = 120 × 3.7 (1)
- 440 N (1)

ALLOW 444 N

REJECT 1176 N (used g for Earth) for the final mark

(b) **[2]**

- 120 kg (1)
- mass does not depend on gravitational field strength OR mass is the amount of matter in the rover, which does not change when it moves (1)

REJECT "mass changes because g is different"

(c) **[2]**

- Jupiter has a much greater mass than the Earth (1)
- the greater the mass of a planet, the stronger its gravitational field OR the greater the gravitational force on each kilogram at its surface (1)

ALLOW gravitational field strength depends on the mass of the planet (and the distance from its centre)

IGNORE references to Jupiter being further from the Sun

Worked solution

(a) The weight equation must be recalled: weight = mass × gravitational field strength. On Mars, W = 120 kg × 3.7 N/kg = 444 N, which is 440 N to 2 significant figures.

(b) Mass is the amount of matter in an object. It is the same everywhere, so the rover still has a mass of 120 kg on the Moon. Only its weight changes, because g on the Moon is different.

(c) Gravitational field strength at the surface of a planet depends on the planet's mass (and on how far the surface is from the centre). Jupiter's mass is much greater than the Earth's, so its gravitational field at its surface is stronger: 25 N/kg compared with 9.8 N/kg.

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10 · Part 1 Paper 1 & 2 Motion in the universe [7]

Mark scheme

(a) **[2]**

Any two from:

- a comet's orbit is highly elliptical (very stretched) but a planet's orbit is almost circular (1)
- a comet's distance from the Sun changes a lot, but a planet's distance stays nearly the same (1)
- a comet's speed changes a lot, but a planet's speed is almost constant (1)
- a comet's orbit can be at a large angle to the plane in which the planets orbit (1)

ALLOW comets usually have much longer time periods than (most) planets (1)

(b) **[2]**

- the comet travels much further from the Sun OR its orbit is much larger, so it has a much greater distance to travel in one orbit (1)
- it moves very slowly when it is far from the Sun (1)

(c) **[3]**

- the comet's distance from the Sun changes (because its orbit is elliptical) (1)
- as the comet moves towards the Sun, the Sun's gravitational force pulls it forwards and makes it speed up OR it gains kinetic energy as it loses gravitational potential energy (1)
- it moves fastest when it is closest to the Sun and slowest when it is furthest from the Sun (1)

ALLOW as it moves away from the Sun, gravity slows it down

Worked solution

(a) Planets move in nearly circular orbits, so their distance from the Sun and their speed hardly change. Comets move in highly elliptical orbits: they come very close to the Sun and then travel far out beyond the planets. Comet orbits are also often tilted at large angles to the orbits of the planets.

(b) The orbit of Halley's Comet takes it much further from the Sun than the Earth, so the path is much longer. In the distant part of its orbit it moves slowly, so one orbit takes a long time.

(c) The Sun's gravitational force always pulls the comet towards the Sun. When the comet is moving inwards, this pull speeds it up (gravitational potential energy changes to kinetic energy). When the comet is moving outwards, the pull slows it down. So the comet is fastest near the Sun and slowest far away.

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11 · Part 1 Paper 1 & 2 Motion in the universe [6]

Mark scheme

(a) **[1]**

- 92.5 × 60 = 5550 s (1)

(b) **[3]**

- substitution: orbital speed = (2 × π × 6.78 × 10^6) / 5550 (1)
- 7680 (1)
- m/s (1)

ALLOW 7700 m/s OR 7.7 × 10^3 m/s OR 7676 m/s

ECF from (a)

(c) **[2]**

- the ISS is further from the centre of the Earth than the Earth's surface is (1)
- gravitational field strength decreases as the distance from the centre of the Earth increases (1)

IGNORE "there is no gravity in space"

Worked solution

(a) Each minute is 60 s, so 92.5 × 60 = 5550 s.

(b) orbital speed = (2 × π × orbital radius) / time period = (2 × π × 6.78 × 10^6 m) / 5550 s. The top line is 4.26 × 10^7 m (the distance round one orbit). Dividing by 5550 s gives 7676 m/s, which is 7680 m/s to 3 significant figures.

(c) Gravitational field strength is not the same everywhere. It gets smaller the further you are from the centre of the planet. The ISS is about 400 km above the surface, so g there is a little smaller (8.7 N/kg) than at the surface (9.8 N/kg). It is not zero: this gravitational force is what keeps the ISS in orbit.

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12 · Part 1 Paper 1 & 2 Motion in the universe [8]

Mark scheme

(a) **[2]**

- 365 × 24 × 60 × 60 (1)
- = 31 536 000 s OR 3.15×10^7 s (1)

(b) **[3]**

- substitution: orbital speed = $(2 \times \pi \times 1.5 \times 10^{11}) / (3.2 \times 10^7)$ (1)

ALLOW 3.15×10^7 s used for the time period

- 2.9×10^4 to 3.0×10^4 (1)
- m/s (1)

(c) **[3]**

- time period = (2 × π × orbital radius) / orbital speed OR $(2 \times \pi \times 2.3 \times 10^{11}) / (2.4 \times 10^4)$ (1)
- = 6.0×10^7 s (1)
- $(6.0 \times 10^7) / 86\,400$ = 700 days (1)

ALLOW 690 to 700 days

ECF for the final mark from an incorrect time period in seconds, correctly converted to days

Worked solution

(a) One day has 24 hours, each hour has 60 minutes and each minute has 60 s. So 365 days = 365 × 24 × 60 × 60 = 31 536 000 s = 3.15×10^7 s, which is about 3.2×10^7 s.

(b) orbital speed = $(2 \times \pi \times 1.5 \times 10^{11}$ m) / $(3.2 \times 10^7$ s). The distance round the orbit is 9.4×10^{11} m, so the speed is 2.9×10^4 m/s (or 3.0×10^4 m/s using 3.15×10^7 s).

(c) Rearrange the equation: time period = (2 × π × orbital radius) / orbital speed = $(2 \times \pi \times 2.3 \times 10^{11}$ m) / $(2.4 \times 10^4$ m/s) = 6.02×10^7 s. One day is 24 × 60 × 60 = 86 400 s, so the time period is $6.02 \times 10^7 / 86\,400$ = 697 days, about 700 days. Check: Mars is further from the Sun and moves more slowly than the Earth, so its year should be longer than 365 days, and it is.

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13 · Part 1 Paper 1 & 2 Motion in the universe [7]

Mark scheme

(a) **[4]**

- time period = 24 × 60 × 60 = 86 400 s (1)
- rearrangement: orbital radius = (orbital speed × time period) / (2 × π) (1)
- substitution: (3.07 × 10^3 × 86 400) / (2 × π) (1)
- 4.2 × 10^7 m (1)

ALLOW 4.22 × 10^7 m

- award 3 marks for 7.0 × 10^5 m (time period left in minutes) OR 1.2 × 10^4 m (time period left in hours) with correct rearrangement

(b) **[1]**

- 4.22 × 10^7 − 6.4 × 10^6 = 3.6 × 10^7 m (1)

ALLOW 3.58 × 10^7 m

ECF from (a)

(c) **[2]**

- the gravitational force of the Earth on the satellite (1)
- this force acts towards the centre of the Earth and continually changes the direction of the satellite's motion, so it moves in a curved path (1)

ALLOW without this force the satellite would move off in a straight line

REJECT "the satellite's engines push it round"

Worked solution

(a) First change the time period into seconds: 24 × 60 × 60 = 86 400 s. Rearrange orbital speed = (2 × π × r) / T to give r = (v × T) / (2 × π). Then r = (3.07 × 10^3 m/s × 86 400 s) / (2 × π) = 2.65 × 10^8 / 6.28 = 4.22 × 10^7 m.

(b) The orbital radius is measured from the centre of the Earth, so subtract the Earth's radius: 4.22 × 10^7 m − 0.64 × 10^7 m = 3.58 × 10^7 m, about 3.6 × 10^7 m.

(c) The Earth's gravitational force pulls the satellite towards the centre of the Earth all the time. The satellite is moving sideways, so this pull keeps bending its path into a circle instead of letting it fly off in a straight line.

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14 · Part 1

Paper 1 & 2 Motion in the universe [4]

Mark scheme

(a) **[1]**

- the Milky Way (1)

(b) **[2]**

- a large collection of stars (1)
- billions of stars (1)

(c) **[1]**

- planet, solar system, galaxy, universe (1)
- all four must be in the correct order

Worked solution

(a) Our solar system is in the Milky Way galaxy.

(b) A galaxy is a large collection of billions of stars. The Sun is just one of the stars in the Milky Way.

(c) A planet orbits the Sun and is part of the solar system. The solar system is a tiny part of the Milky Way galaxy. The universe is a large collection of billions of galaxies. So the order is planet, solar system, galaxy, universe.

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15 · Part 1 Paper 1 & 2 Stellar evolution [5]

Mark scheme

(a) **[3]**

Any three from:

- a nebula is a cloud of dust and gas (mainly hydrogen) (1)
- gravitational force pulls the dust and gas together OR the cloud collapses (1)
- the material becomes hotter (and denser) as it is pulled together (1)
- when it is hot enough, nuclear fusion starts and the star begins to give out energy (1)

(b) **[2]**

- gravitational force pulls the material of the star inwards (1)
- this is balanced by an outward force OR pressure due to the energy released by nuclear fusion (1)

ALLOW the inward and outward forces are balanced (1 mark only if neither force is identified)

Worked solution

(a) A star begins as a nebula, a huge cloud of dust and gas. Gravitational attraction pulls the material closer together, so the cloud collapses and heats up. When the centre is hot and dense enough, hydrogen nuclei fuse together, releasing energy, and a star is born.

(b) Gravity is always trying to make a star collapse. In a main-sequence star, the energy released by fusion in its core creates an outward pressure. The two effects are balanced, so the star stays the same size and is stable for billions of years.

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16 · Part 1 Paper 1 & 2 Stellar evolution [5]

Mark scheme

(a) **[3]**

Any three from:

- the hydrogen in the core runs out (1)
- the Sun expands to become a red giant (1)
- its outer layers drift away into space (1)
- the core that is left contracts to form a white dwarf (1)

ALLOW the white dwarf cools down (and fades) over time

(b) **[2]**

- white (1)

ALLOW blue-white OR blue

- the Sun is now yellow, and a star with a higher surface temperature is white or blue OR the hotter the surface of a star, the further its colour moves from red towards blue (1)
- the reason must link colour to surface temperature

Worked solution

(a) When the Sun uses up the hydrogen in its core, it swells to become a red giant: it gets much bigger and its surface cools, so it looks red. Later its outer layers drift off into space and the hot core that is left shrinks to become a white dwarf, a small, dense, very hot star that slowly cools.

(b) A star's colour depends on its surface temperature. The Sun is a yellow star today. A star with a much higher surface temperature is white (or blue-white), which is why the remnant is called a white dwarf.

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17 · Part 1 Paper 1 & 2 Stellar evolution [5]

Mark scheme

(a) **[2]**

- Betelgeuse (1)
- it has the lowest surface temperature, and red stars are the coolest stars (1)

(b) **[2]**

- the higher the surface temperature, the more the colour changes from red towards blue OR hotter stars are blue and cooler stars are red (1)
- correct order of colours with increasing temperature: red, orange, yellow, white, blue (1)

ALLOW any three of these colours in the correct order for the second mark

(c) **[1]**

- redder (than the Sun) (1)

ALLOW more red OR more orange OR less blue OR closer to red

REJECT bluer OR whiter

Worked solution

(a) Red stars are the coolest stars. Betelgeuse has the lowest surface temperature in the table (3500 K), so it is the red star.

(b) A star's colour is related to its surface temperature. As the surface temperature increases, the colour changes from red to orange, yellow, white and then blue. Astronomers can therefore classify stars by colour, and the colour tells them roughly how hot the surface is.

(c) Arcturus has a lower surface temperature (4300 K) than the Sun (5800 K). Cooler stars are redder, so Arcturus appears redder than the Sun (in fact it looks orange, between the red Betelgeuse and the yellow Sun).

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18 · Part 1 Paper 2 only Stellar evolution [6]

Mark scheme

(a) **[2]**

- a measure of the brightness of a star (1)
- as it would appear from a standard distance (1)

ALLOW standard distance given as 10 parsecs OR 32.6 light-years

REJECT "how bright the star looks from the Earth" for the second mark

(b) **[2]**

- Deneb (1)
- its absolute magnitude (−8.4) is lower OR more negative than that of Sirius (+1.4) (1)
- no mark for Deneb without a reason based on absolute magnitude

(c) **[2]**

- Sirius is much closer to the Earth (8.6 light-years compared with 2600 light-years) (1)
- the brightness seen from the Earth depends on distance as well as on how much light the star gives out OR light from a star spreads out, so a more distant star appears dimmer (1)

Worked solution

(a) How bright a star looks from the Earth depends on how far away it is. To compare stars fairly, astronomers use absolute magnitude: the brightness a star would have if it were at a standard distance (10 parsecs).

(b) On the magnitude scale, a lower (more negative) number means a brighter star. Deneb has an absolute magnitude of −8.4, much lower than Sirius's +1.4, so Deneb really gives out much more light.

(c) Sirius is only 8.6 light-years away, but Deneb is about 2600 light-years away. Light spreads out as it travels, so the much more distant Deneb looks dimmer from the Earth even though it gives out far more light.

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19 · Part 1 Paper 1 & 2 Stellar evolution [6]

Mark scheme

(a) **[4]**

Any four from:

- the star expands to become a red supergiant (1)

ALLOW red giant

- it explodes as a supernova (1)
- its outer layers are thrown out into space (1)
- the core that is left collapses (1)
- the core forms a neutron star OR a black hole (1)

ALLOW black hole for the most massive stars and neutron star for the others

(b) **[2]**

- star Y has a much greater mass than the Sun (1)
- so the gravitational force on its core is much greater and the core collapses much further, to form a neutron star or a black hole instead of a white dwarf (1)

ALLOW only stars with a large mass become supergiants and explode as a supernova (for the second mark)

Worked solution

(a) When a massive star like Y runs out of hydrogen in its core, it swells to become a red supergiant. It then explodes in a supernova, which throws its outer layers out into space. The core that is left collapses. It becomes an extremely dense neutron star, or, if the mass left is large enough, a black hole.

(b) The mass of a star decides how it ends. The Sun does not have enough mass: it becomes a red giant and then a white dwarf, and never explodes. Star Y has about 20 times the mass, so the gravitational forces are much stronger. Its core collapses much further, giving a supernova and then a neutron star or black hole.

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20 · Part 1 Paper 2 only Cosmology [5]

Mark scheme

(a) **[2]**

- the frequency heard is lower than 700 Hz (1)
- the wavelength heard is longer than the wavelength emitted (1)

(b) **[2]**

- the ambulance moves away from the observer between emitting one wave and the next OR each wave is emitted from a position further from the observer (1)
- so the waves (compressions) reaching the observer are further apart, giving a longer wavelength (1)

(c) **[1]**

- the driver is moving with the siren OR there is no relative motion between the siren and the driver (1)

Worked solution

(a) When a source of waves moves away from an observer, the observer receives waves with a longer wavelength and a lower frequency than those emitted. So the observer hears a lower note than 700 Hz.

(b) The siren keeps emitting waves while it moves away. Each new wave starts from a point a little further from the observer than the last one, so the waves are stretched out. The distance between compressions, the wavelength, is longer, and fewer waves arrive each second.

(c) The driver moves along with the siren, so the distance between the driver and the siren does not change. There is no relative motion, so the driver hears the true frequency of 700 Hz.

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21 · Part 1 Paper 2 only Cosmology [6]

Mark scheme

(a) **[1]**

- 669 − 656 = 13 nm (1)

(b) **[3]**

- substitution: 13 / 656 = velocity / (3.0×10^8) (1)
- rearrangement: velocity = (13 / 656) × 3.0×10^8 (1)
- 5.9×10^6 m/s (1)

ALLOW 5.95×10^6 m/s OR 6.0×10^6 m/s

ECF from (a)

(c) **[2]**

- away from the Earth (1)
- the wavelength received is longer than the wavelength measured in the laboratory OR the light is red-shifted (1)
- no mark for "away" without a reason

Worked solution

(a) change in wavelength = observed wavelength − reference wavelength = 669 nm − 656 nm = 13 nm.

(b) change in wavelength / reference wavelength = 13 / 656 = 0.0198. This equals velocity / speed of light, so velocity = 0.0198 × 3.0×10^8 m/s = 5.9×10^6 m/s. The units of wavelength cancel in the ratio, so it does not matter that they are in nm.

(c) The wavelength received (669 nm) is longer than the wavelength in the laboratory (656 nm). The light is shifted towards the red end of the spectrum, which happens when the source is moving away from the observer.

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22 · Part 1 Paper 2 only Cosmology [6]

Mark scheme

(a) **[4]**

- change in wavelength / reference wavelength = $(2.4 \times 10^7) / (3.0 \times 10^8)$ (1)
- = 0.080 (1)
- change in wavelength = 0.080 × 486 = 38.9 nm (1)
- wavelength received = 486 + 38.9 = 525 nm (1)

ALLOW 524.9 nm

- award 3 marks for 447 nm (change in wavelength subtracted)

(b) **[2]**

- change in wavelength = 0.080 × 434 = 34.7 nm (1)
- wavelength received = 434 + 34.7 = 469 nm (1)

ALLOW 468.7 nm

ECF from the ratio found in (a)

Worked solution

(a) First find the ratio: velocity / speed of light = $(2.4 \times 10^7) / (3.0 \times 10^8) = 0.080$. This is also change in wavelength / reference wavelength, so change in wavelength = 0.080 × 486 nm = 38.9 nm. The galaxy is moving away, so the light is red-shifted and the wavelength gets longer: 486 + 38.9 = 524.9 nm, about 525 nm.

(b) Every line in the light from galaxy H is shifted by the same fraction, 0.080, of its own wavelength. change in wavelength = 0.080 × 434 nm = 34.7 nm, so the wavelength received is 434 + 34.7 = 468.7 nm, about 469 nm. Check: 469 / 434 = 1.08, the same as 525 / 486.

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23 · Part 1 Paper 2 only Cosmology [6]

Mark scheme

(a) **[2]**

- the greater the distance of the galaxy, the greater the red-shift (1)
- correct use of data, e.g. when the distance roughly doubles from 120 to 250 million light-years, the red-shift roughly doubles from 0.009 to 0.019 OR red-shift is (approximately) proportional to distance (1)

(b) **[4]**

Any four from:

- red-shift means the light received has a longer wavelength, so the galaxies are moving away from the Earth (1)
- (almost) all distant galaxies show a red-shift, so galaxies are moving away in every direction (1)
- more distant galaxies have a greater red-shift (1)
- so more distant galaxies are moving away faster (1)
- so the galaxies are moving apart from each other OR the space between galaxies is increasing, which means the universe is expanding (1)

Worked solution

(a) Each galaxy that is further away has a larger red-shift. The values go up in step: about 0.0075 for every 100 million light-years, so the red-shift is roughly proportional to the distance.

(b) A red-shift means the wavelength of the light has increased, which happens when the source is moving away. Light from distant galaxies in every direction is red-shifted, so they are all moving away from us. The more distant a galaxy is, the bigger its red-shift, so the faster it is moving away. If all the galaxies are moving apart, and the furthest ones are moving fastest, the space between them must be getting bigger: the universe is expanding.

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24 · Part 1 Paper 2 only Cosmology [5]

Mark scheme

(a) **[2]**

Any two from:

- it is radiation detected from all directions in space (1)
- it has almost the same intensity in every direction (1)
- it is radiation left over from the early universe, shortly after the Big Bang (1)
- the word "microwave" alone does not score

(b) **[3]**

Any three from:

- soon after the Big Bang the universe was extremely hot (and dense) (1)
- so it gave out high-energy, short-wavelength radiation (1)
- as the universe expanded (and cooled), the wavelength of this radiation was stretched OR the radiation was red-shifted (1)
- so the radiation is now detected as microwaves (1)

Worked solution

(a) CMB radiation is microwave radiation that reaches the Earth from every direction in space, with almost the same intensity in each direction. It is the remains of radiation from the very early universe.

(b) The Big Bang theory says the early universe was extremely hot and dense, so it was filled with high-energy radiation of short wavelength. As the universe expanded, this radiation was stretched to longer and longer wavelengths. Today it has reached the microwave region, and it is found everywhere, just as the theory predicts.

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25 · Part 1 Paper 2 only Cosmology [5]

Mark scheme

(a) **[3]**

Any three from:

- the universe began from a single, extremely small point (1)
- it was extremely hot and dense (1)
- it has been expanding ever since (1)
- it has cooled as it has expanded (1)

ALLOW galaxies and stars formed later as the universe cooled

ALLOW about 13.8 billion years ago

(b) **[2]**

- the wavelength will increase OR become longer (1)
- because as the universe expands, the radiation is stretched further OR it is red-shifted more (1)

ALLOW the universe will continue to cool, so the radiation will have a longer wavelength

Worked solution

(a) According to the Big Bang theory, the universe started from one extremely small, extremely hot and dense point. Since then it has been expanding, and as it expanded it cooled. As it cooled, matter came together to form stars and galaxies.

(b) The CMB radiation started with a short wavelength. Expansion of the universe stretches the radiation, which is why it is now in the microwave region. If the universe keeps expanding, the radiation will keep being stretched, so its wavelength will become even longer.

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